

Bent, Bypassed, Unbowed: US AI-Chip Export Controls and China’s Compute

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July 2026

analysis pass of record: 2026-07, **natex** v0.2.0, seeds 7 (DiD) and 20260716 (DiK battery; estimator deterministic)

Abstract

The October 2023 US export controls on AI accelerators bent China’s *legally* imported AI-chip stock sharply downward. A group difference-in-kinks (DiK) contrast of China’s legal H100-equivalent stock against the four US hyperscalers at the 2023-10-17 rule gives -0.00154 ln-units/day (HC1 se 0.00047; ≈ -0.56 ln-units/yr), is negative in 25 of 25 estimated specifications (defensible band -0.3 to -0.56 ln-units/yr), is null at the October-2022 loophole-round placebo cutoff, and its placebo-treated group is clean at defensible bandwidths. The effect on China’s *total* chip stock — including smuggled chips — is roughly half as large and statistically fragile (-0.00076 , $t = -1.54$, 95% CI spanning zero): smuggled Nvidia chips in China grew from 27.8k H100e in Q1 2024 to 662k by Q4 2025, doubling every 4.7 months, and now exceed China’s legal Nvidia stock $1.65\times$. Finally, a country-level difference-in-differences on the count of $\geq 10^{24}$ -FLOP training runs finds no suppression at all: China’s big-run count rose $+3.2$ runs/quarter *relative to* every control-group counterfactual after the October-2022 round — but that estimate fails placebo-in-space (exact $p = 1.0$; every pseudo-treated unit shows a larger studentized effect), so it is descriptive, not causal. The controls demonstrably bent the legal supply channel; substitution through smuggling and domestic chips roughly halved and de-robustified the effect on China’s actual compute stock; and no slowdown in China’s large training runs is detectable in these data.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security imposed the first broad export controls on advanced AI accelerators and chipmaking tools bound for China, a policy contemporaries immediately read as a landmark attempt to choke off China’s access to frontier AI compute [1]. The rule’s performance-and-interconnect thresholds left room for compliant workarounds — Nvidia’s A800 and H800 — and on 17 October 2023 a second round closed that loophole, banning the workaround chips outright. Whether such controls work is an old question in trade policy: export controls impose costs on the controlling country’s own firms and reroute rather than stop trade when substitutes exist [3, 13]. Firm-level evidence from the earlier entity-list controls finds large collateral costs — US suppliers to targeted Chinese firms lose customers, market value (roughly \$130 billion), lending, and employment, with broad supply-chain decoupling but no reshoring [6].

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On the evasion margin, [9] warned in 2023 that AI-chip smuggling into China, then likely in the hundreds to low thousands of units, would industrialize; by 2026, diversion-and-resale estimates put smuggled compute at hundreds of thousands of H100-equivalents [12]. What this literature lacks is a quasi-experimental estimate of whether the controls actually bent China’s compute accumulation and large-scale training activity.

This paper provides three complementary tests — three legs of one story — using two Epoch AI datasets [8, 7] and the `natex` toolkit [14]. Leg 1 is a country-level difference-in-differences (DiD) on the quarterly count of $\geq 10^{24}$ -FLOP training runs, with a scan-based localization step [11]. Leg 2 is a group difference-in-kinks (DiK) — the design of [2], building on the regression kink design of [5] — contrasting the growth-rate bend in China’s legal chip stock against US hyperscalers at the 2023-10-17 rule, with placebo cutoffs in the spirit of [10]. Leg 3 repeats the DiK on China’s total stock including smuggled chips and quantifies the smuggling substitution directly. The verdict is deliberately mixed and stated as such: Leg 2 is credible, Leg 3 is attenuated and fragile, Leg 1 is descriptive only.

2 Data

Country panel (Leg 1). From Epoch AI’s notable-models database [8] we build a panel of 4 country groups (US, China, EU, Other) $\times$ 26 quarters (2020Q1–2026Q2), $n = 104$ cells. The outcome is the count of training runs $\geq 10^{24}$ FLOP per quarter; a secondary outcome is the median $\log_{10}$ training compute. Models are assigned to countries by the lead (first-listed) organization, so multinational collaborations are attributed to the lead country. The declared treatment is `china_controlled`: China from 2022Q4 (the first control round) onward. Two caveats bind: 2026 quarters undercount compute-known runs ($\sim 13\%$ FLOP coverage, a reporting lag), and counts measure quantity of big runs, not frontier capability.

Chip stocks (Legs 2–3). The chip-stock series come from Epoch AI’s AI Chip Owners data [7]: quarterly cumulative H100-equivalent (H100e) stock by designer. From these we form ln cumulative stock series for: China’s legal stock; China’s total stock including the smuggled Nvidia series [12]; the four US hyperscalers (Amazon, Microsoft, Google, Meta) as the primary control; neoclouds (Oracle, CoreWeave, xAI) as an alternative control; and an “Other” group used as a placebo-treated group. Each series covers 16 quarters from 2022-03-31; the incomplete 2026-03-31 quarter is dropped (8 rows whose “cumulative” totals decrease). The running variable is days since 2023-10-17. The smuggled series begins 2024-03-31 and is itself part of the treatment response — which is exactly why Legs 2 and 3 are reported side-by-side.

3 Design and Methods

Leg 1: SuDDDS DiD. The `natex` sector-by-time DiD scan (SuDDDS) searches subsets of units and break dates maximizing a likelihood-ratio scan statistic [11], calibrated by permutation ($q = 99$, seed 7, windows $\{4, 7, 13\}$). We run it three ways: (i) known-treatment mode (Bernoulli model, `wcc`) on the declared treatment; (ii) a normal-model cross-check with a dependence-preserving `ar1_unit` null; (iii) discovery mode with no treatment column, asking whether the export-control dates would be found autonomously. Effects for China at 2022Q4 come from DD, synthetic-control, and GESS estimators, followed by a manual placebo battery (placebo-in-space with the Abadie china-removed convention, placebo-in-time at 2021Q3).

Legs 2–3: group difference-in-kinks. In [2] the DiK contrasts the same schedule kink across two *time periods*. Here the export-license schedule bends *for China only* at a common date, so the Böckerman contrast is taken across *groups* at one cutoff: $\text{DiK} = (\text{China slope kink}) - (\text{hyperscaler slope kink})$, each kink estimated by local linear regression on each side of the cutoff (right-minus-left), triangular kernel, primary bandwidth 548 days. Identification is parallel-kink: absent the controls, China’s growth bend at Oct-2023 would have matched the hyperscalers’; the control group differences out the global supply-curve bend (the hyperscalers’ own kink is $-0.000666/\text{day}$). Robustness: bandwidths $\{365, 548, 730\}$ days $\times$ $\{\text{triangular, uniform}\}$ kernels, 45-day donuts, six placebo cutoffs (including the Oct-2022 first round at -375 days — the A800/H800 loophole-round anticipation contrast), and a placebo-treated group (Other vs hyperscalers, both untreated), in the spirit of [10]. Leg 3 re-runs the identical design on China’s total stock including smuggled chips.

Honest-inference notes, stated as run. The pipeline’s refusals and trivial passes are reported, not patched over. Leg 1: the built-in `tau_randomization_test` correctly refuses ($p = \text{NaN}$, 3 non-treated units < 5 usable placebos) rather than fabricating a p -value, so the manual placebo-in-space battery is the inference of record; the scan $p = 0.010$ is the $q = 99$ permutation floor; with 4 units, all panel p -values are descriptive. Leg 2: HC1 standard errors on 6 quarterly points per cell overstate precision on a cumulative-stock series; CR1 clustering by calendar year keeps significance but has $G = 4$ clusters with $t(G - 1)$ critical values — a known few-cluster over-rejection regime [4]; the sign-stability across 25 specifications is the real evidence. The density-kink-difference test passes trivially (both groups share an identical quarterly grid; estimate exactly 0.0, p reported as null), no predetermined covariates exist in a two-group quarterly aggregate (the placebo-treated group and the Oct-2022 placebo round substitute for the covariate-kink battery), and no Fieller set is produced (sharp design, known unit denominator). A recorded `natex` UX finding: the CLI aliases the group contrast through its time interface (`--time china --t0 1`), so output cells are labeled `pre_*/post_*` and its caveat text misdescribes the cross-group contrast as a “time-stable marginal response.”

4 Results

Leg 1: localization succeeds, attribution fails. In known-treatment mode SuDDDS recovers the declared treatment exactly: top subset $\{\text{China}\}$, $T_0 = 11 = 2022\text{Q4}$, window $W = 13$, Bernoulli LLR 6.3715, panel-null $p = 0.010$ (the $q = 99$ floor); the composition check passes ($p = 1.0$) and the anticipation test passes (Holm $p = 1.0$). The normal-model cross-check under the dependence-preserving `ar1_unit` null finds the mirror image — the complement $\{\text{EU, Other, US}\}$ at $T_0 = 13$ (2023Q2), LLR 14.9487 — but $p = 0.14$: a 4-unit panel cannot separate a persistent step from unit-level AR(1). Discovery mode does *not* find the controls: the top hit is a $\{\text{China, US}\}$ *downshift* at 2025Q4 (LLR 11.277, $p = 0.13$; runner-ups ALL@2025Q4, LLR 7.232, and $\{\text{China, US}\}@2024\text{Q2}$, LLR 5.835), with no export-control date in the top five — and the 2025Q4 downshift is at least partly the reporting-lag artifact. Effect estimates for China at 2022Q4 (Table 1, panel A) are uniformly *positive*: DD $\hat{\tau} = +3.214$ runs/qtr (se 1.157, $t = 2.779$, 95% CI $[+0.947, +5.481]$), synthetic control +3.933, GESS +3.842, DD at the 2023Q4 round +4.455, and DD on median $\log_{10}$ compute +0.649 $[+0.322, +0.976]$. Taken literally: no suppression — China’s big-run count *outgrew* every control-implied counterfactual. But the placebo battery kills any causal reading of the magnitude: pseudo-treating each control unit at the same date yields studentized $|t|$ of 5.37 (EU), 2.86 (Other), and 5.16 (US) against China’s 2.78 (china-removed convention) — exact placebo-in-

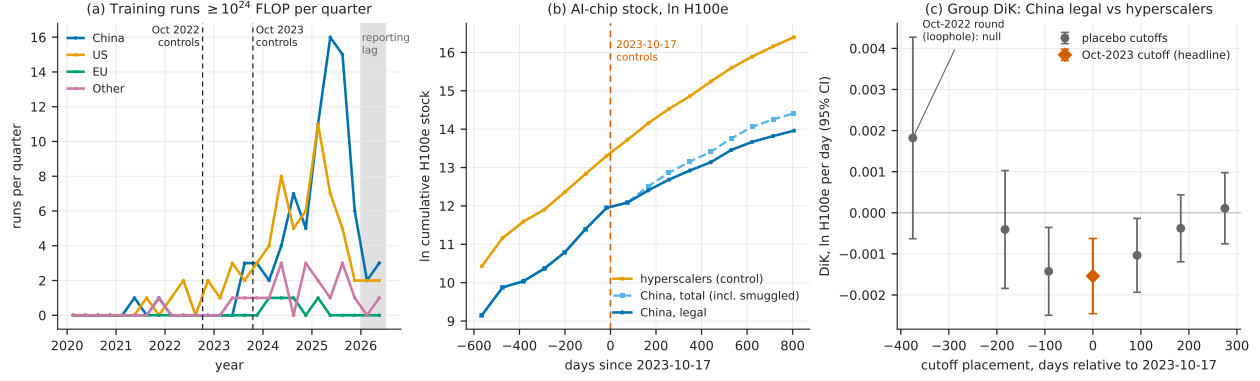


Figure 1: (a) Leg 1: quarterly count of $\geq 10^{24}$ -FLOP training runs by country group, with both export-control rounds marked; the shaded region is the $\sim 13\%$ -coverage reporting-lag zone. The panel is a two-speed world: US and China accelerate, EU and Other stagnate. (b) Legs 2–3: In cumulative H100e chip stock for the hyperscaler control group, China’s legal stock, and China’s total stock including smuggled chips, around the 2023-10-17 cutoff. (c) Leg 2: group-DiK estimates (bandwidth 548 days, triangular kernel) at the true cutoff (diamond) and at six placebo cutoff placements, with 95% HC1 intervals; the Oct-2022 loophole round at -375 days is null.

space $p = 4/4 = 1.0$. Placebo-in-time is clean (China at 2021Q3: $\tau = -0.30$, $t = -1.33$), and the built-in randomization test correctly refused ($p = \text{NaN}$, $3 < 5$ placebos). The robust statement is qualitative only: China’s large-run count did not fall behind its counterfactual after either control round.

Leg 2: the legal channel credibly bent. The headline group DiK (China legal vs hyperscalers, 2023-10-17, bandwidth 548 days, triangular kernel) is -0.0015404 ln-units/day (HC1 se 0.000467, $t = -3.298$, 95% CI $[-0.002456, -0.000625]$), i.e. ≈ -0.56 ln-units/yr of growth-rate change; with CR1 clustering by year, se 0.000299, $t = -5.152$ ($G = 4$, few-cluster caveat). The cell kinks are -0.000666 /day (hyperscalers) vs -0.002206 /day (China legal). The estimate is negative in **25 of 25** estimated specifications: -0.000869 to -0.002026 across $\text{bw}365/548/730 \times \text{triangular/uniform}$ (t from -2.07 to -6.57), and $-0.0018996/-0.0012108/-0.0008011$ on the 45-day-donut grid (t $-3.92/-2.04/-1.60$). The placebo-cutoff battery (bw548; empirical size 2/6) localizes the kink to within one quarter of the rule: the Oct-2022 first round at -375 days is *null* at $+0.0018203$ (se 0.001251, $p = 0.146$) — the loophole round produced no bend, a pseudo-pre-trend pass — and -183d ($p = 0.578$), $+183\text{d}$ ($p = 0.364$), $+275\text{d}$ ($p = 0.805$) are null, while -92d ($p = 0.0089$) and $+92\text{d}$ ($p = 0.0240$) reject: with quarterly data, one-quarter shifts still straddle the same bend, which still brackets 2023-10-17. The placebo-treated group (Other vs hyperscalers, both untreated) is null at $\text{bw}548/730$ ($t = +1.34$, -0.29 , -0.20 , -0.75) but rejects at $\text{bw}365$ (triangular $+0.000958$, $t = 2.49$; uniform $+0.001219$, $t = 3.04$) — parallel-kink fails among untreated groups at four-points-per-cell bandwidths, so the $\text{bw}365$ cells are discounted and the defensible band is $\text{bw}548-730$: -0.0009 to -0.0015 /day (≈ -0.3 to -0.56 ln-units/yr).

Leg 3: smuggling halves and de-robustifies the effect. On China’s total stock including smuggled chips, the same design gives -0.0007638 (se 0.000496, $t = -1.540$), with a 95% CI of $[-0.001736, +0.000209]$ that spans zero — roughly half the legal-channel bend, and t ranges from -0.55 to -3.08 across the six specifications (the only rejection is at the discounted $\text{bw}365$). The

Table 1: Headline estimates across the three legs.

Specification	$\hat{\tau}$	se	t	note
<i>Panel A: Leg 1 — DiD, China at 2022Q4 (runs $\geq 10^{24}$ FLOP per quarter)</i>				
DD (control = other 3 units)	+3.214	1.157	2.78	CI [+0.947, +5.481]
Synthetic control	+3.933	—	—	CI [+1.533, +6.334]
GESS	+3.842	—	—	CI [+1.442, +6.243]
DD at 2023Q4 round	+4.455	—	—	CI [+1.719, +7.190]
DD, median $\log_{10}$ compute	+0.649	—	—	CI [+0.322, +0.976]
placebo-in-space exact $p = 4/4 = 1.0 \Rightarrow$ descriptive, not causal				
<i>Panel B: Leg 2 — group DiK, $\ln$ H100e per day, China legal vs hyperscalers</i>				
bw548 triangular (headline)	-0.001540	0.000467	-3.30	HC1
bw548 triangular, CR1 by year	-0.001540	0.000299	-5.15	$G = 4$ clusters
bw548 uniform	-0.001132	0.000525	-2.16	
bw730 triangular / uniform	-0.001140 / -0.000869		-2.85 / -2.07	
bw365 triangular	-0.002026	0.000308	-6.57	discounted
placebo cutoff -375d (Oct-2022)	+0.001820	0.001251		$p = 0.146$: null
<i>Panel C: Leg 3 — group DiK, China total stock (incl. smuggled) vs hyperscalers</i>				
bw548 triangular	-0.000764	0.000496	-1.54	CI [-0.001736, +0.000209]
bw730 triangular / uniform	-0.000458 / -0.000235		-1.10 / -0.55	
bw365 triangular	-0.001049	0.000340	-3.08	discounted

Notes: Panel A: **natex** DiD effect estimators on the 4-country $\times$ 26-quarter panel (seed 7); 95% CIs in the note column. Panels B–C: sharp group difference-in-kinks, right-minus-left slope contrasts in $\ln$ cumulative H100e stock per day at the 2023-10-17 cutoff; HC1 standard errors unless noted; bandwidths in days; “discounted” marks bw365 cells where the placebo-treated group (Other vs hyperscalers) rejects. Multiply DiK estimates by 365.25 for annual growth-rate changes.

mechanism is visible in the raw series: smuggled Nvidia compute in China grew from 27.8k H100e (Q1 2024) to 662k H100e (Q4 2025), doubling every 4.7 months — the fastest-growing series in the dataset — and now stands at $1.65\times$ China’s legal Nvidia stock. Even so, China’s share of the world chip stock roughly halved to 8.7% over the period despite absolute growth: the controls did not stop China’s accumulation, but China fell behind a world that accelerated faster.

5 Caveats and Conclusion

Four caveats bound the claims. First, Leg 1’s panel has four units — the floor for any panel placebo inference — and its placebo-in-space failure is informative precisely because it shows the units are not exchangeable (a two-speed world), so no DiD counterfactual built from EU/Other/US is credible for China’s magnitude; every Leg-1 p -value is descriptive. Second, Leg 2’s inference rests on 6 quarterly points per cell: HC1 overstates precision on a cumulative series, CR1 has $G = 4$ clusters [4], and the ± 92 -day placebo rejections show the kink is localized only to ± 1 quarter — the design cannot distinguish 2023-10-17 from an adjacent-week event, though no rival same-quarter candidate of comparable relevance is known to us. Third, the smuggled series underlying Leg 3 is itself an estimate with wide uncertainty [12] and begins only in Q1 2024, endogenously to the treatment. Fourth, counts of big training runs measure quantity, not frontier capability; controls aimed at chip access can bind on quality and efficiency margins invisible to Leg 1’s outcome.

The three-leg verdict: *credible* on the legal channel (Leg 2) — the October-2023 controls bent China’s legal chip-stock growth down by roughly 0.3–0.56 $\ln$ -units/yr relative to every control group, with the loophole-round placebo null exactly where the policy history says it should be; *attenuated and fragile* on total compute (Leg 3) — smuggling substitution roughly halves the point

estimate and removes its robustness; and *descriptive only* on training activity (Leg 1) — China’s count of large training runs did not fall behind any counterfactual, but the design cannot certify that as a causal null. Export controls, on this evidence, bent the channel they legally govern, were substantially bypassed in aggregate, and left China’s large-run training activity unbowed.

Reproducibility. All estimates were produced with `natex` v0.2.0 [14]: the Leg-1 runs at seed 7 ($q = 99$) and the Leg-2/3 battery at seed 20260716 (the DiK estimator itself is deterministic), from frozen extracts of the Epoch AI datasets [8, 7] (not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline DD and DiK estimates against the numbers of record before drawing.

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